

ANALOG & DIGITAL CIRCUITS

Lecture 3

Computer Science Department, TUCN

Analog Circuits

Operational Amplifiers (Opamps)

DC Power Supplies

Oscillators

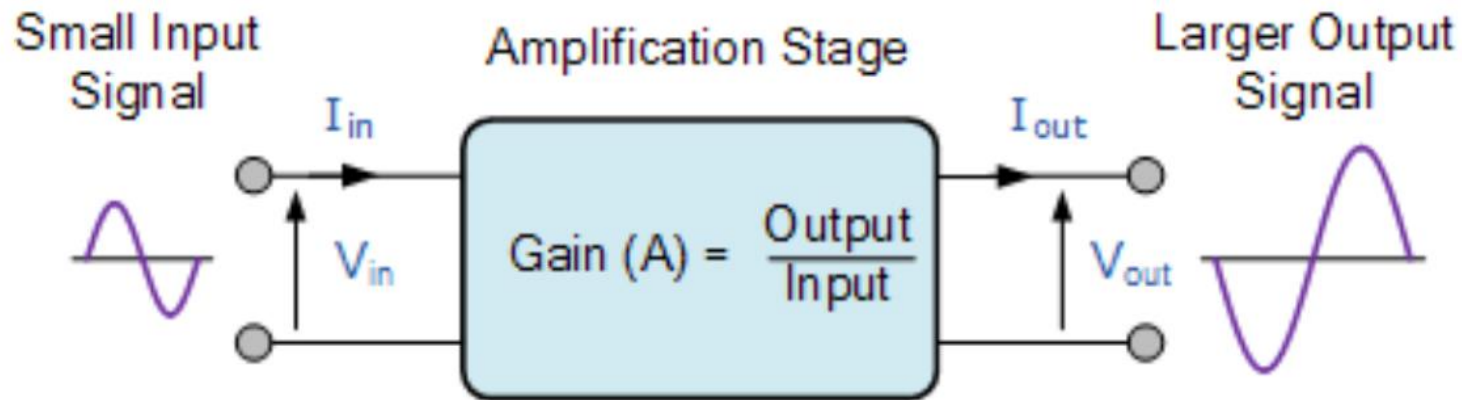
Amplifiers

- electronic device / circuit used to **increase the magnitude** of the input signal



$V_{OUT} = A * V_{IN} \Rightarrow$ linear relationship $\Rightarrow$ linear amplifier

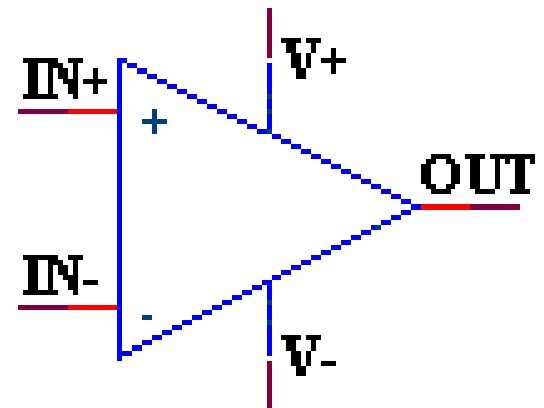
- no distortions introduced this way



Operational Amplifiers

- High-gain differential amplifier, using voltage feedback, providing stabilized voltage gain
- Symbol of differential amplifier
 - inputs
 - Inverting input (IN-)
 - Non-inverting input (IN+)
 - output (OUT)
 - dc power supplies
 - Positive supply (V+)
 - Negative supply (V-)

Intended to work on very small input voltage signals -> to increase the signal's magnitude => larger so we can work with it



Op-amps sense the difference between the voltage signals applied at their input terminals and then amplify it by some pre-determined gain

Characteristics

- Very high input impedance (ideal ∞ - infinite input resistance)
- Very small input currents (ideal 0)
- Very high amplification (voltage gain) (A_v - ideal ∞)
- Output voltage bounded by power supplies values (V_+ and V_-)
- Very small difference between the absolute values of input voltages (ideal 0)
- Very small output impedance (ideal 0 – zero output resistance)
- Two operation modes:
 - differential mode: $V_d = V_{IN+} - V_{IN-}$
 - common mode: $V_c = (V_{IN+} + V_{IN-})/2$

(useful mode for rejecting noise signals);

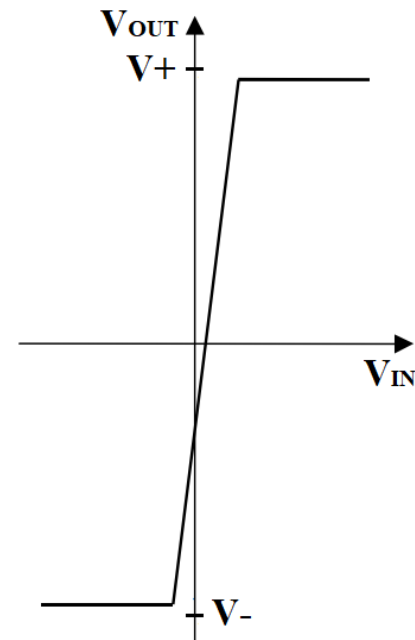
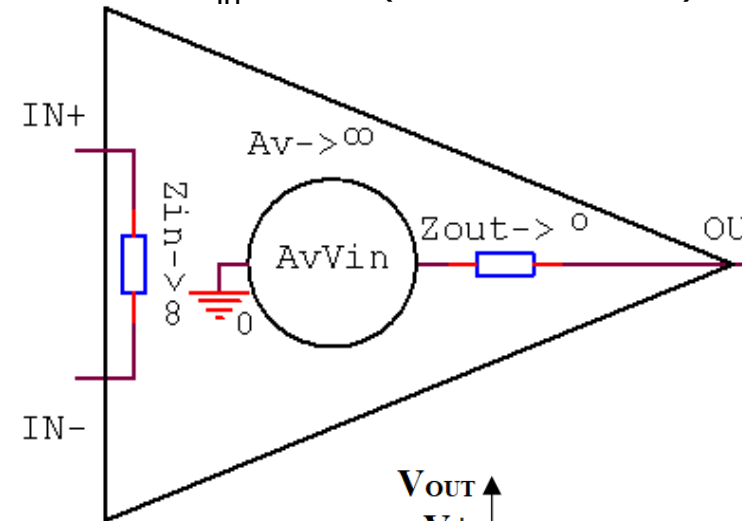
common mode rejection ratio CMRR parameter defined; expressed logarithmically in decibells dB: $CMRR[dB] = 20\log_{10}(A_d/A_c)$

* A_d is the differential gain and A_c is the common mode gain

Assumptions:

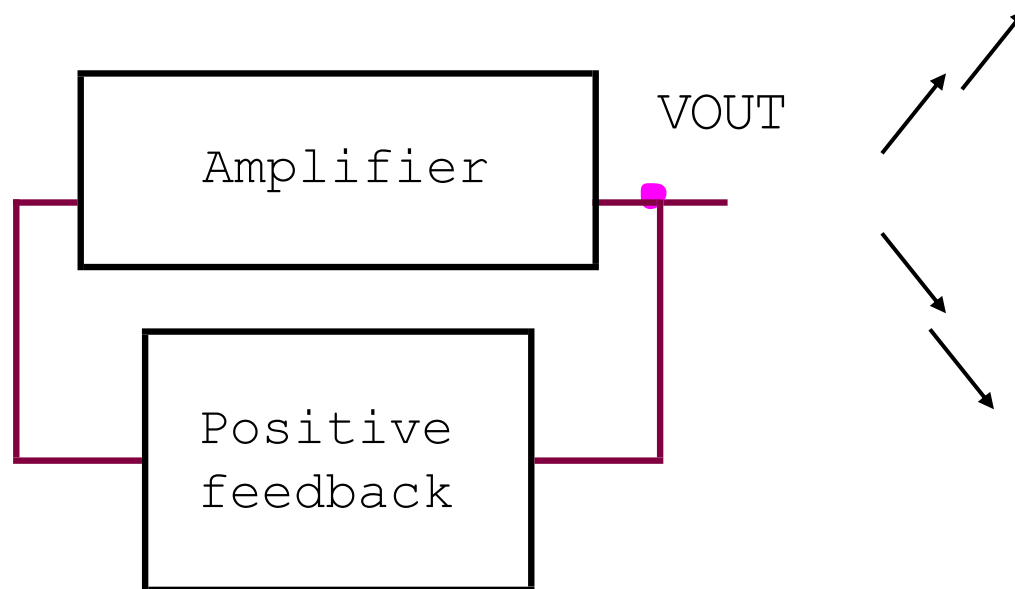
$$I_{in} \sim 0 \text{ amps}$$

$$V_{in} \sim 0 \text{ V } (V_{in+} - V_{in-} = 0)$$



Positive Reaction (feedback)

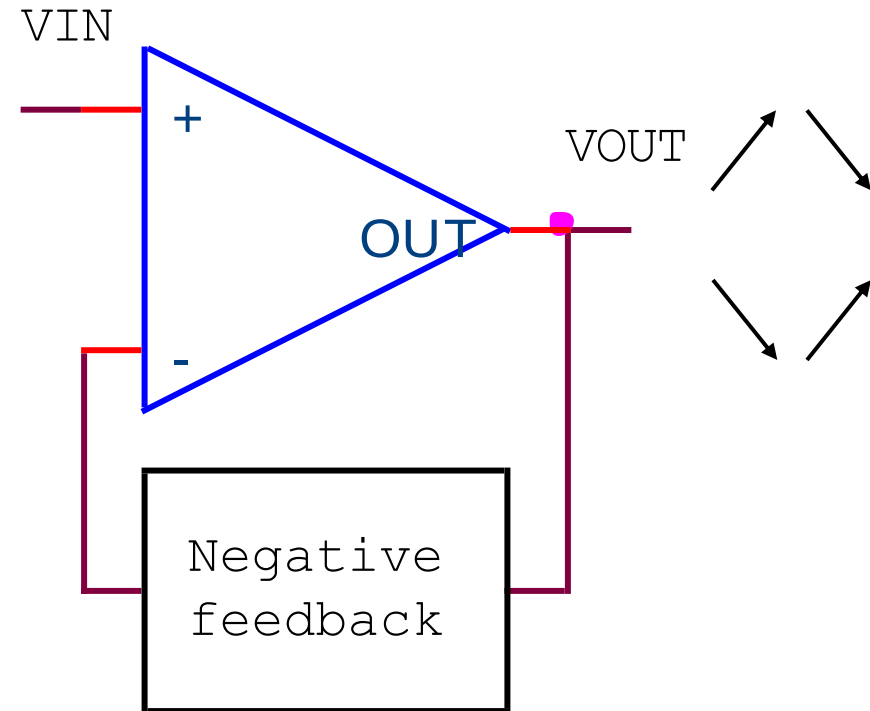
- An amount of the output signal is brought to the input with a total zero phase shift producing an increase of the output signal



$$V_{out} \text{ (at time } t_{N+1}) = V_{in} + V_{out} \text{ (at time } t_N)$$

Negative Reaction (feedback)

- An amount of the output signal is fed back to circuit's input, with a phase shift of 180° from input signal, implying a decrease of the output signal



Constant-Gain OpAmp

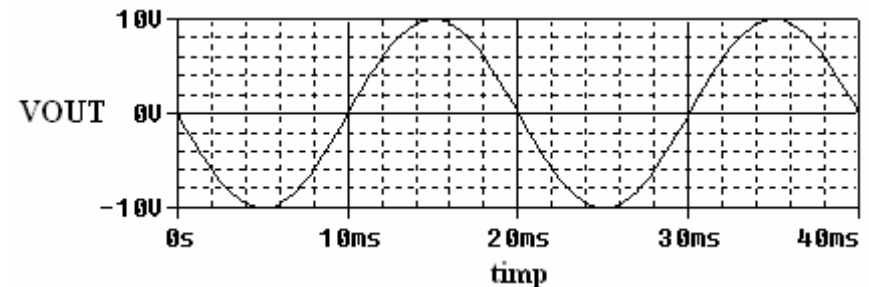
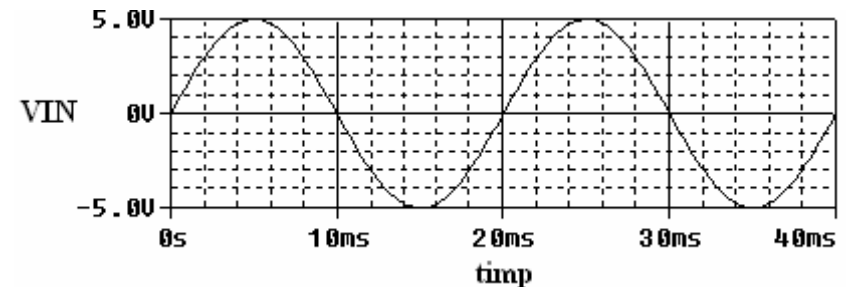
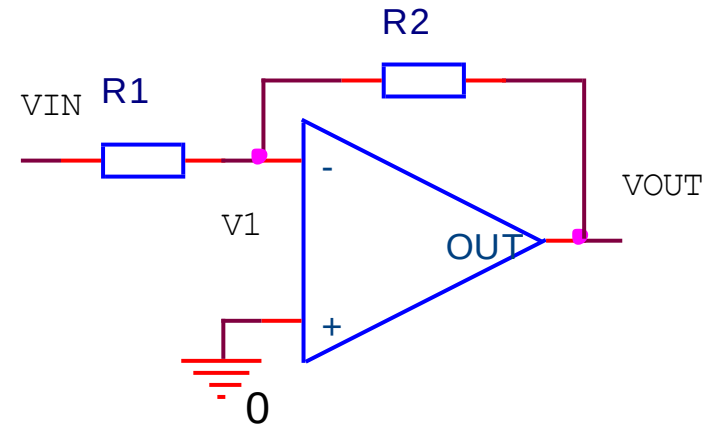
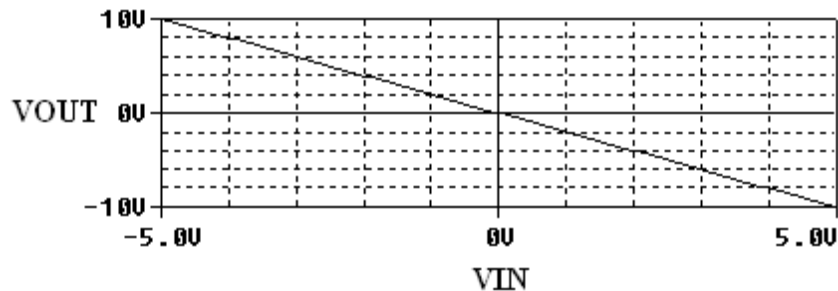
Here, 2 *theoretical* rules

- $V_1 = 0V$
- $I_{R1} = I_{R2}$

1. "No current flows into the input terminal"
2. "V1 always equals V2". (= 0 in our case)

$$I = \frac{V_{IN}}{R_1} = -\frac{V_{OUT}}{R_2}$$

$$V_{OUT} = -\frac{R_2}{R_1} V_{IN}$$

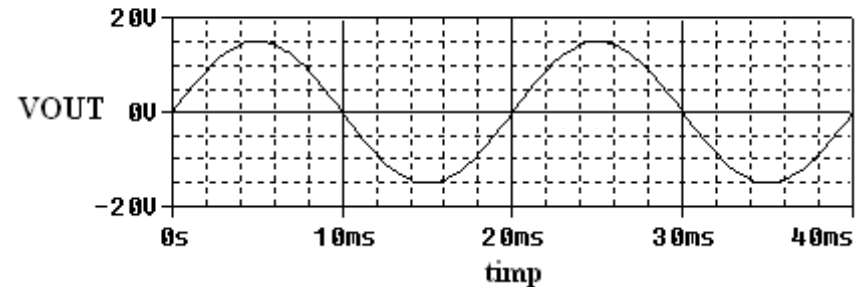
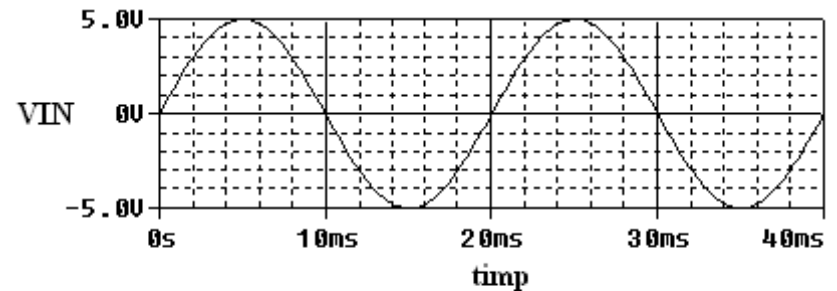
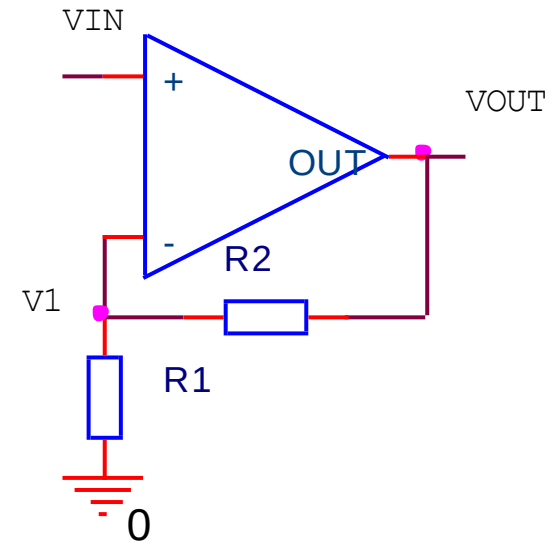
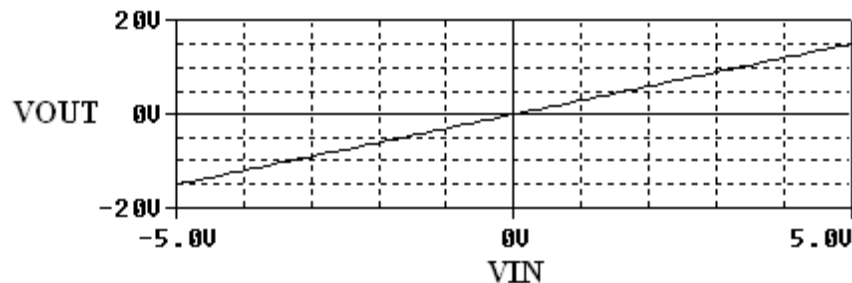


Noninverting OpAmp

- $V_{IN} = V_1$
- $I_{R1} = I_{R2}$

$$V_{IN} = V_1 = \frac{R_1}{R_1 + R_2} V_{OUT}$$

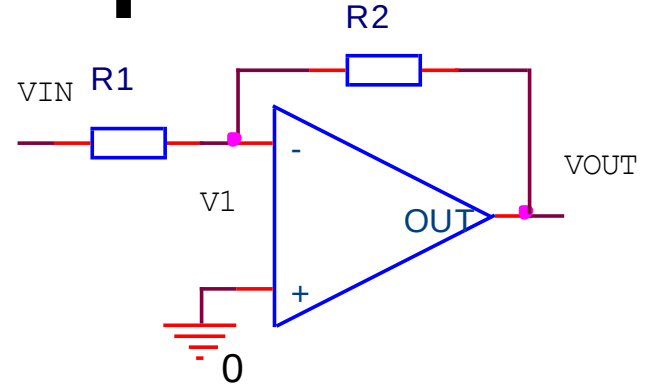
$$V_{OUT} = \left(1 + \frac{R_2}{R_1}\right) V_{IN}$$



Summing OpAmp

Remember the Constant-Gain OpAmp

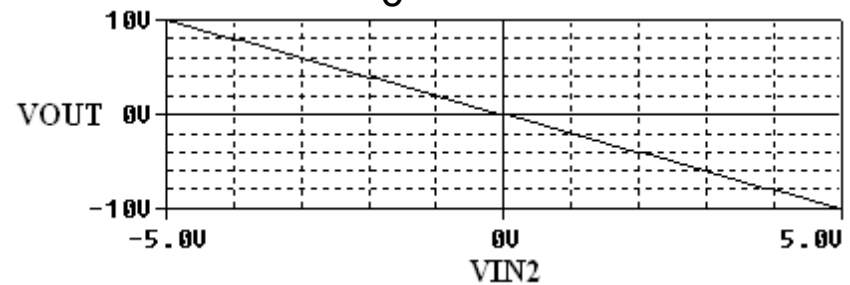
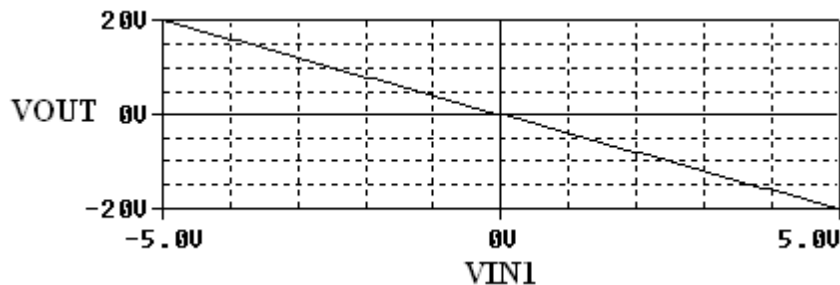
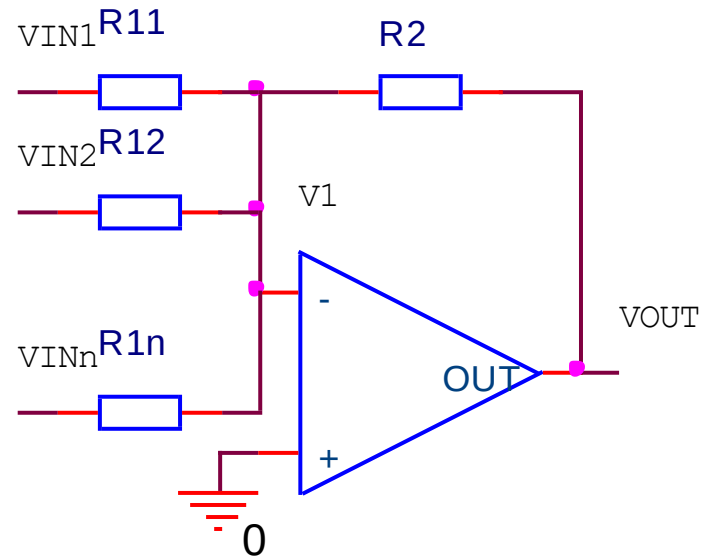
$$V_{OUT} = -\frac{R_2}{R_1} V_{IN} \longrightarrow$$



- $V_1 = 0V$
- $I_{R2} = I_{R11} + \dots + I_{R1n}$

$$I_{R2} = -\frac{V_{OUT}}{R_2} = \frac{V_{IN1}}{R_{11}} + \dots + \frac{V_{INn}}{R_{1n}}$$

$$V_{OUT} = -\left(\frac{R_2}{R_{11}} V_{IN1} + \dots + \frac{R_2}{R_{1n}} V_{INn}\right)$$



Integrator

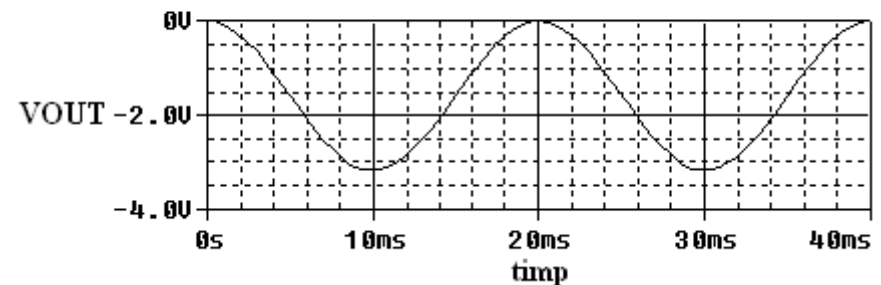
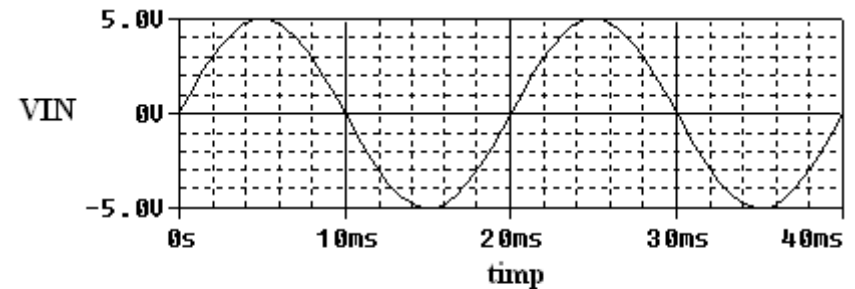
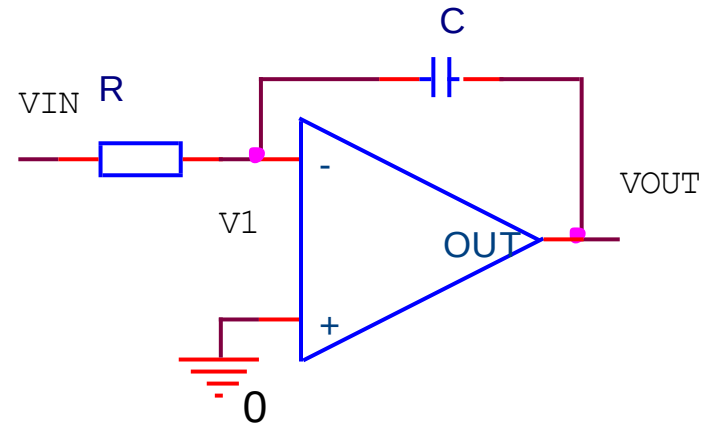
$$Z_C = \frac{1}{j\omega C} = \frac{1}{sC}$$

- $V_1 = 0V$
- $I_R = I_C$

$$I = \frac{V_{IN}}{R_1} = -\frac{V_{OUT}}{Z_C} = -sCV_{OUT}$$

$$V_{OUT} = -\frac{1}{sCR} V_{IN}$$

$$v_{out}(t) = -\frac{1}{RC} \int v_{in}(t) dt$$



Differentiator

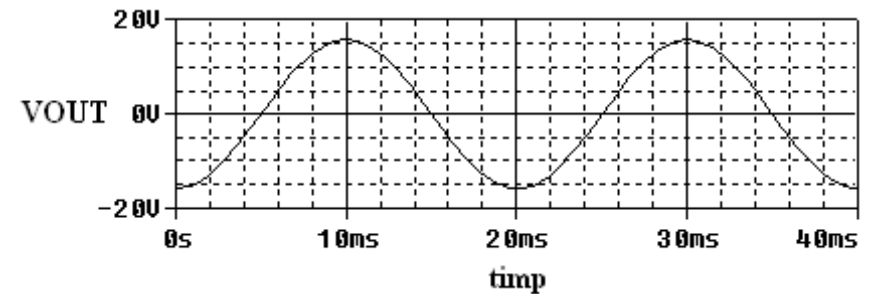
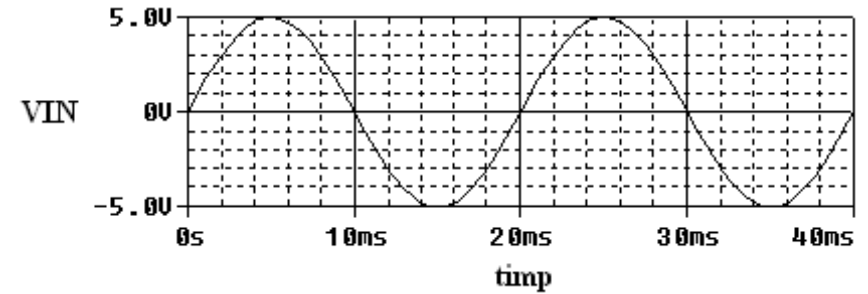
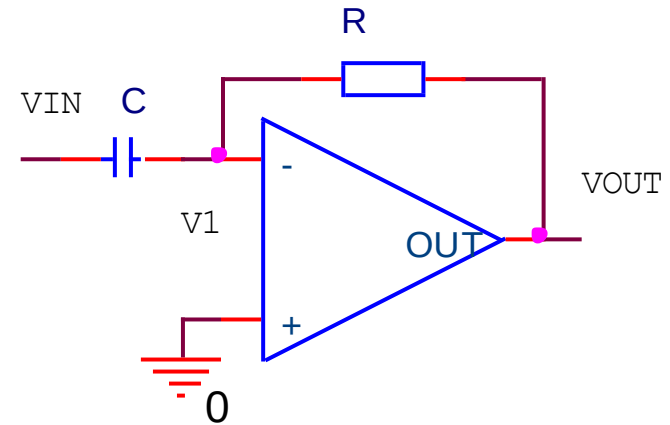
$$Z_C = \frac{1}{j\omega C} = \frac{1}{sC}$$

- $V_1 = 0V$
- $I_R = I_C$

$$I = -\frac{V_{OUT}}{R} = \frac{V_{IN}}{Z_C} = sCV_{IN}$$

$$V_{OUT} = -sCRV_{IN}$$

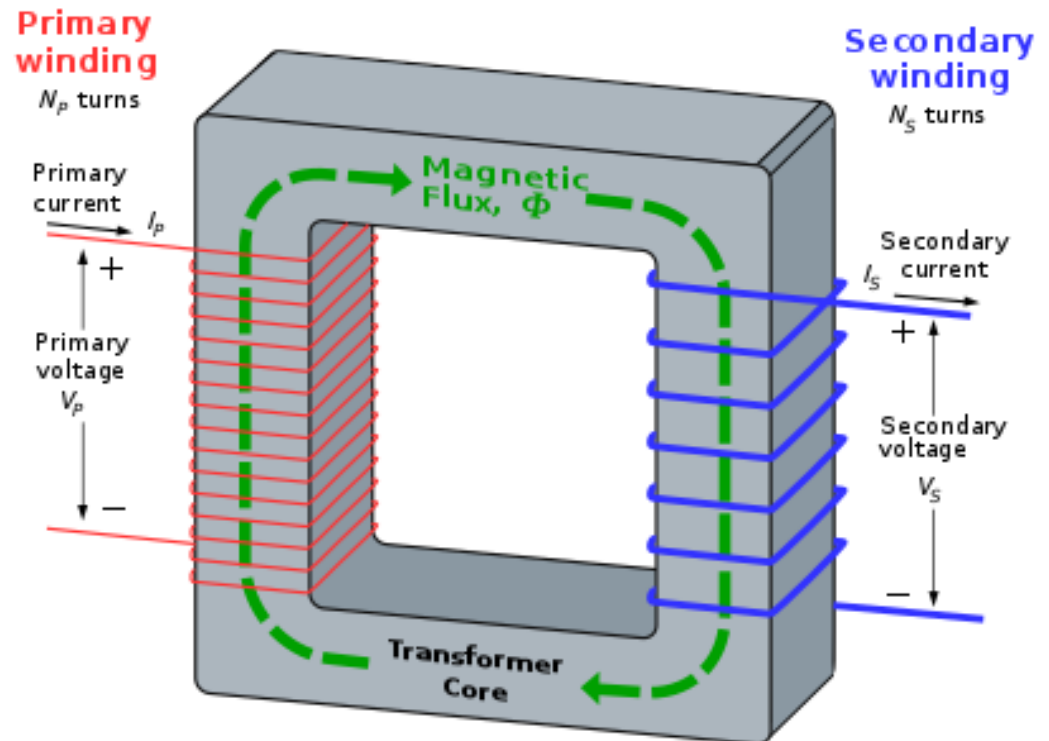
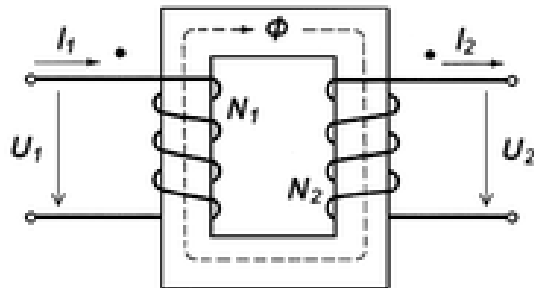
$$V_{out}(t) = -RC \frac{dV_{in}(t)}{dt}$$



Transformers

- separates circuit from ac power supply network and steps input ac voltage to a desired ac level, accordingly to the needed dc voltage level to be obtained

$$\frac{U_1}{U_2} \simeq \frac{E_1}{E_2} = \frac{N_1}{N_2} = k$$



Power Supply Circuits

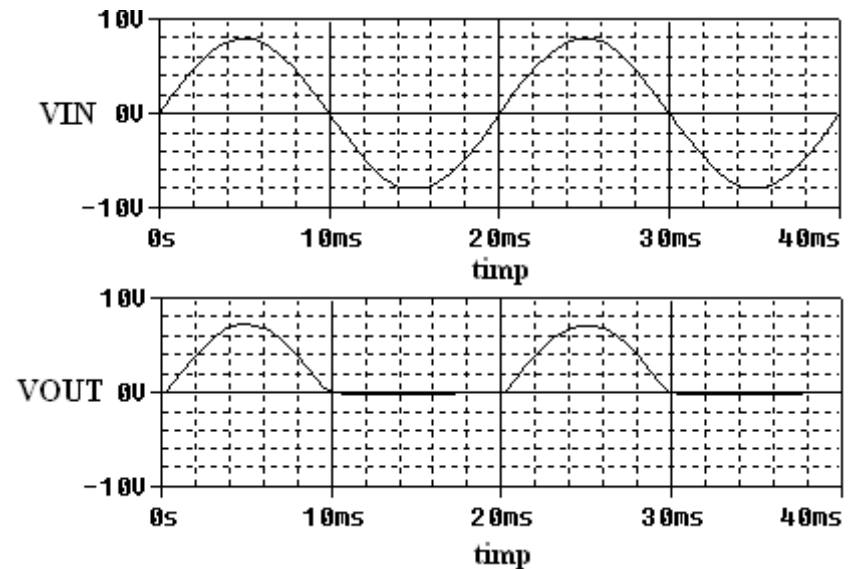
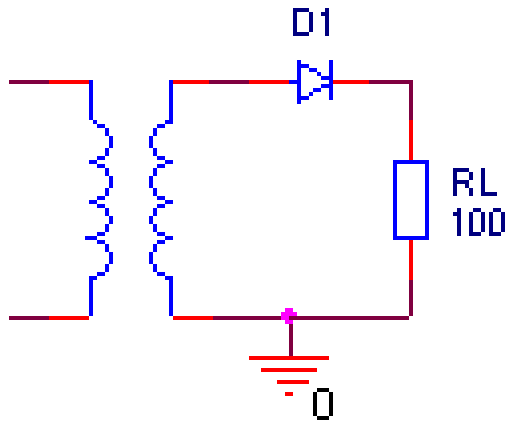
Convert analog supply voltage into desired regulated dc voltage

Circuit consists on:

- transformer: separates circuit from ac power supply network and steps input ac voltage to a desired ac level, conform to the needed dc voltage level to be obtained
- rectifier: to half-wave or full-wave rectify the ac signal, getting a dc component non zero
- filter: attenuation of ac voltage irregularities (as coming from rectifier)
- voltage regulator: regulates the filtered dc voltage to a desired value for output voltage

Half-Wave Rectifier

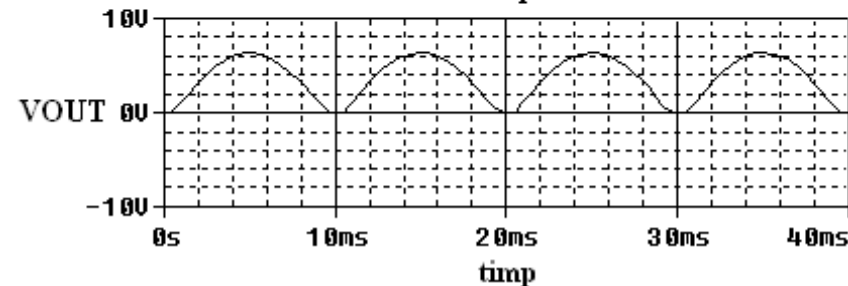
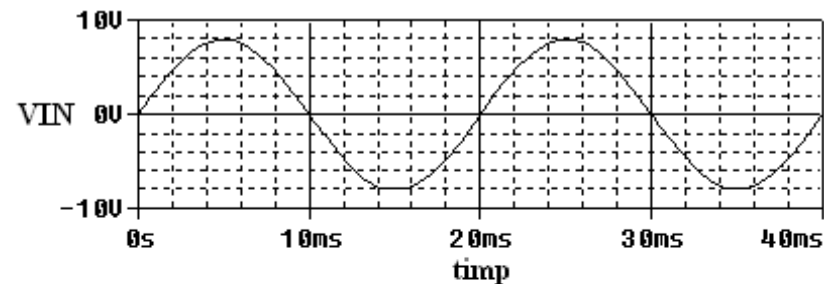
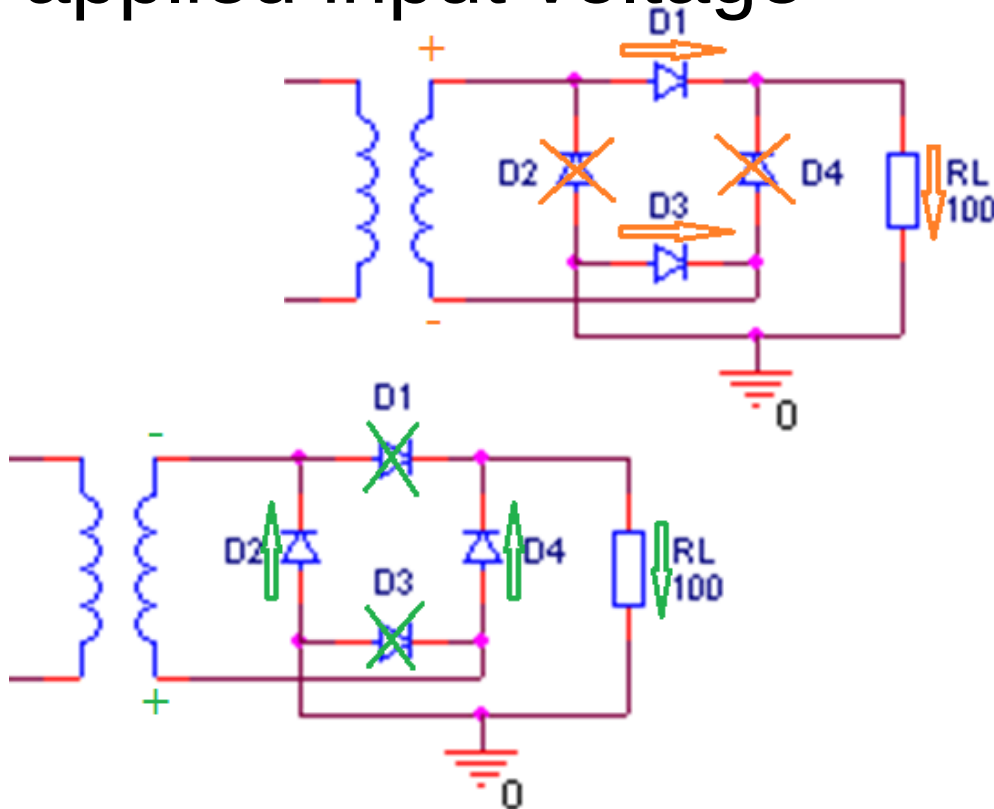
- D_1 is open for positive polarities and off for negative polarities of the applied input voltage



Full-Wave Bridge Rectifier

Four diodes in a bridge configuration

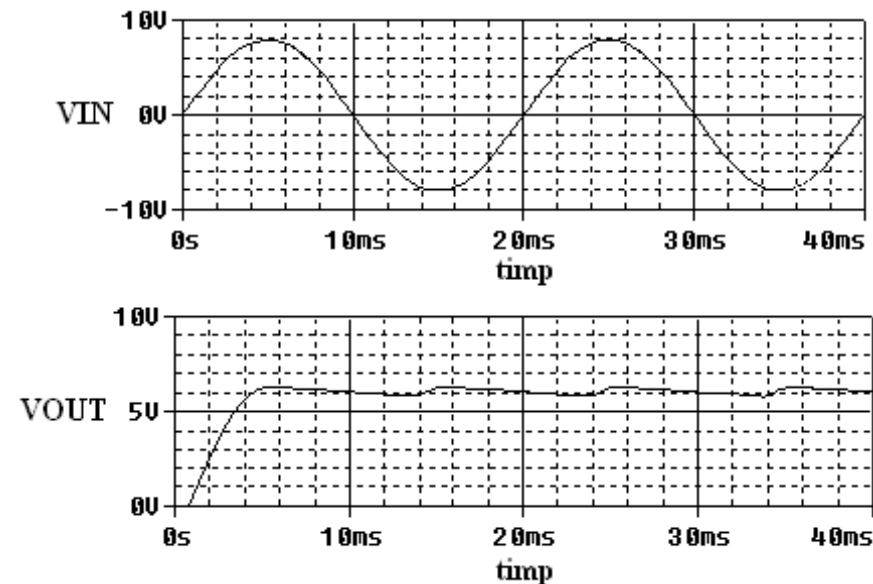
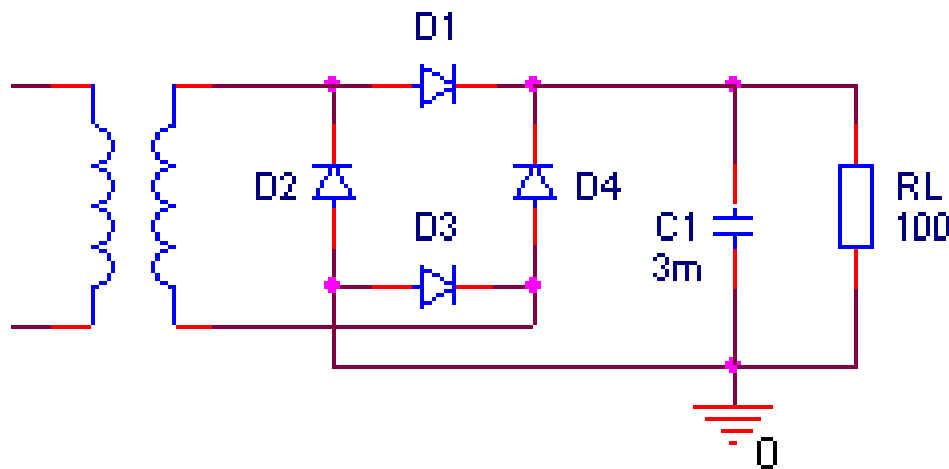
- D_1 and D_3 are open for positive waves
- D_2 and D_4 are open for negative waves of applied input voltage



Rectifier with filtered output

A capacitor C_1 in parallel with load resistance R_L

- If rectifier provides a higher voltage than the voltage drop on capacitance, it stores energy
- If rectifier provides a lower voltage than capacitor's voltage drop, C_1 will generate energy



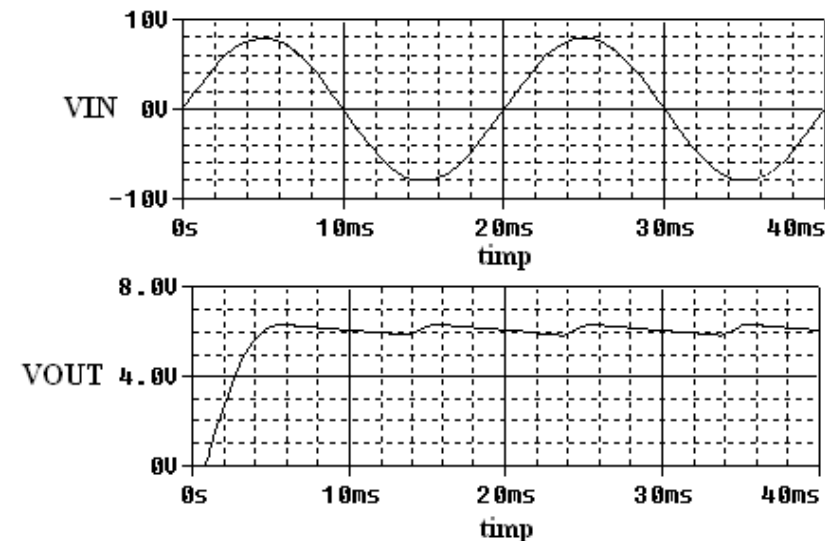
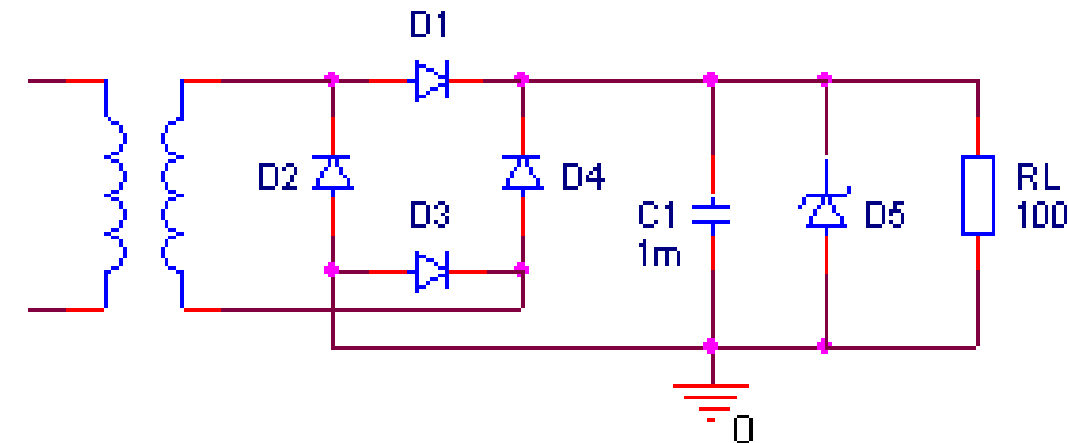
Voltage Regulators

Zener diode shunt regulator

Output voltage must stay fixed, for different loads (R_L values)

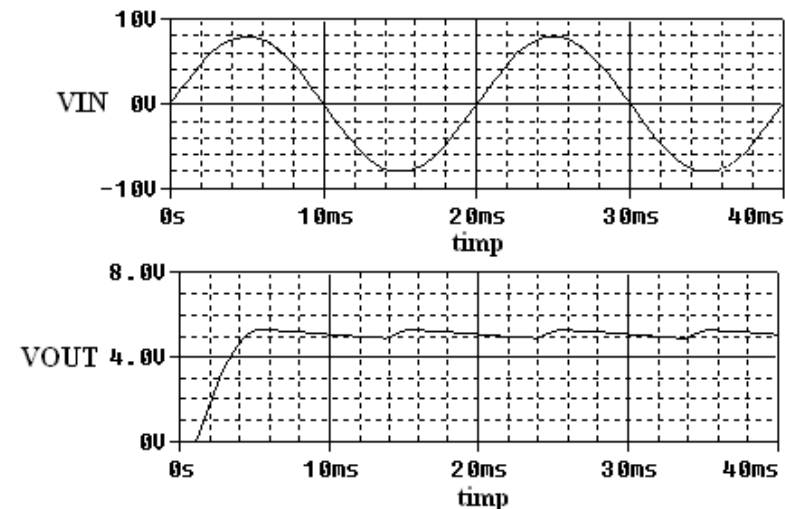
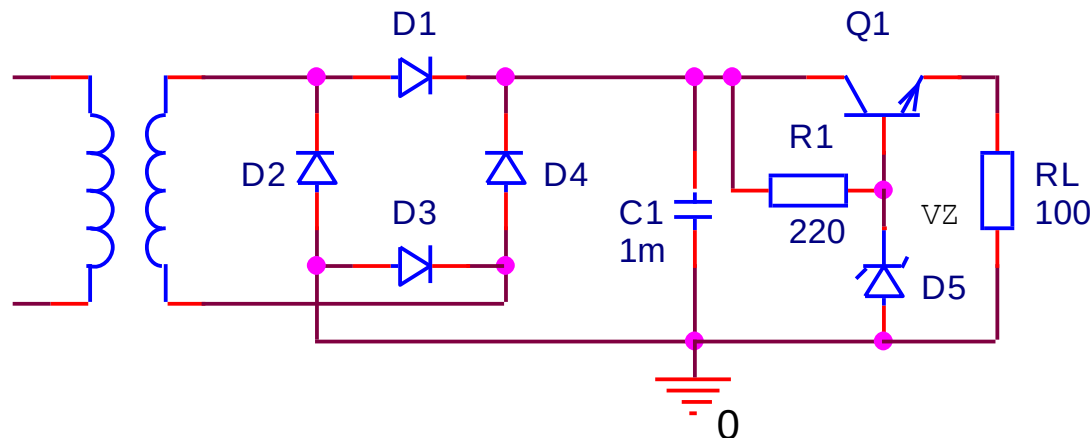
Filtered output voltage has a dc component and some ac variations (ripple voltage)

- Zener diode shunt regulator based on non-linearity of the current-voltage characteristics for a Zener diode
- Allows for high variations of output currents, for small variations of diode's reverse voltage drop



Voltage regulator with negative feedback and without error amplifier

- Negative feedback from collector to base circuit of Q1
- Output voltage (v_{OUT}) compared with a reference voltage (v_Z)
- comparison generates an error signal applied to a control element
- if v_{OUT} increases, control element (***negative reaction***) will impose voltage decrease, and vice-versa
- For transistor Q1: $v_{BE} = v_Z - v_{OUT}$
- if v_{OUT} goes higher $\rightarrow v_{BE}$ lower $\rightarrow v_{CE}$ higher $\rightarrow v_{OUT}$ lower
- if v_{OUT} lower $\rightarrow v_{BE}$ higher $\rightarrow v_{CE}$ lower $\rightarrow v_{OUT}$ higher
- $V_{OUT} = V_Z - V_{BE}$



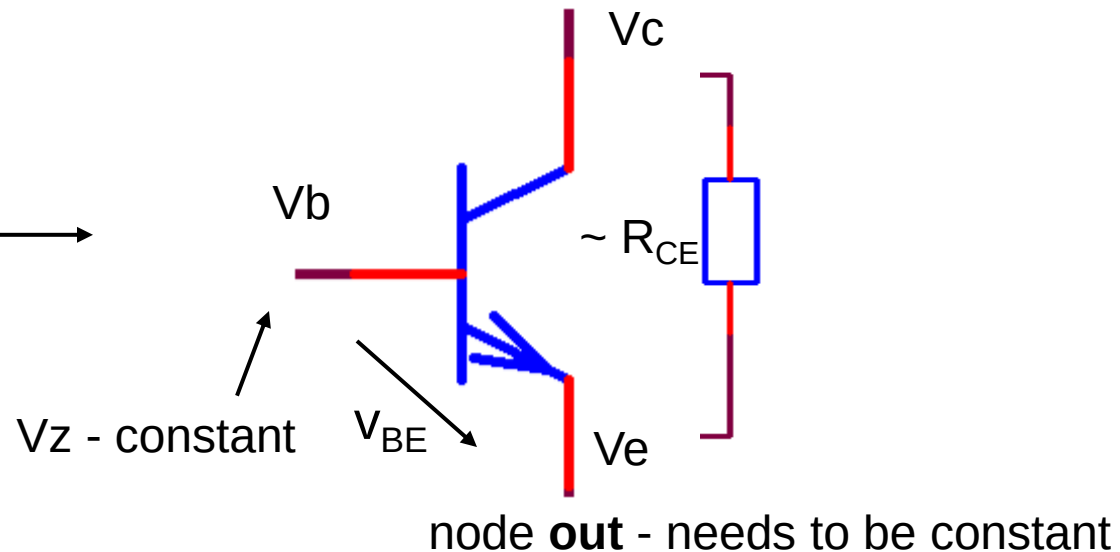
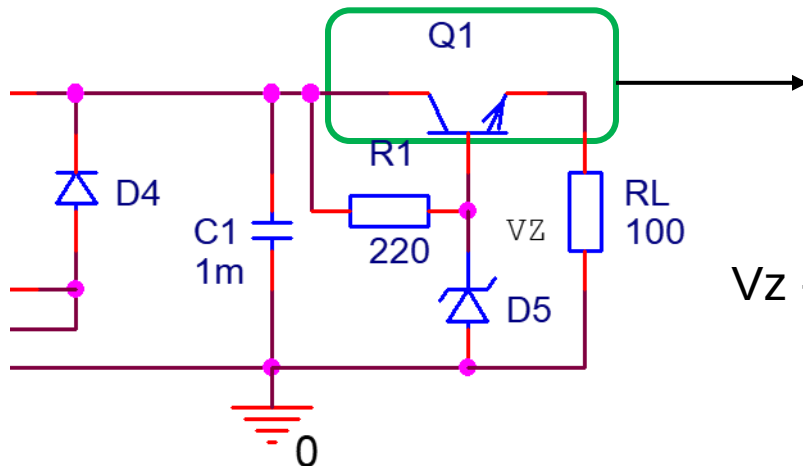
$$V_{BE} = V_Z - V_{OUT}; V_{BE} = 0.7V$$

If Q opened (FAR region), $v_{BE} \uparrow \Rightarrow R_{CE} \downarrow$ between C and E decreases

if v_{OUT} goes higher $\uparrow \rightarrow v_{BE}$ lower $\downarrow \rightarrow R_{CE} \uparrow \rightarrow v_{CE}$ higher $\uparrow \rightarrow v_{OUT}$ lower $\downarrow$

if v_{OUT} lower $\rightarrow v_{BE}$ higher $\rightarrow v_{CE}$ lower $\rightarrow v_{OUT}$ higher

$$V_{OUT} = V_Z - V_{BE}$$

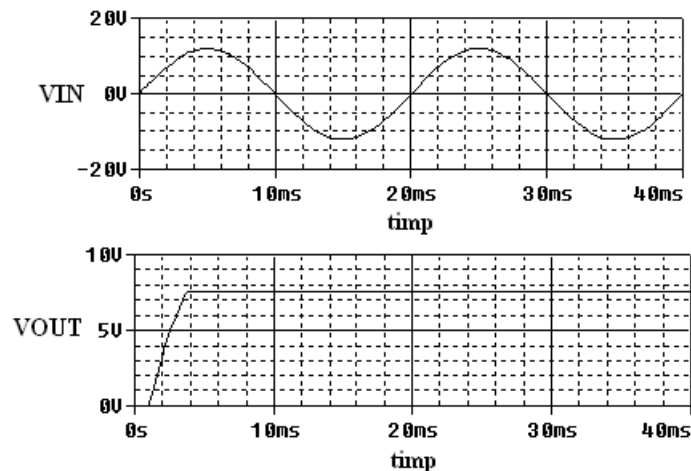
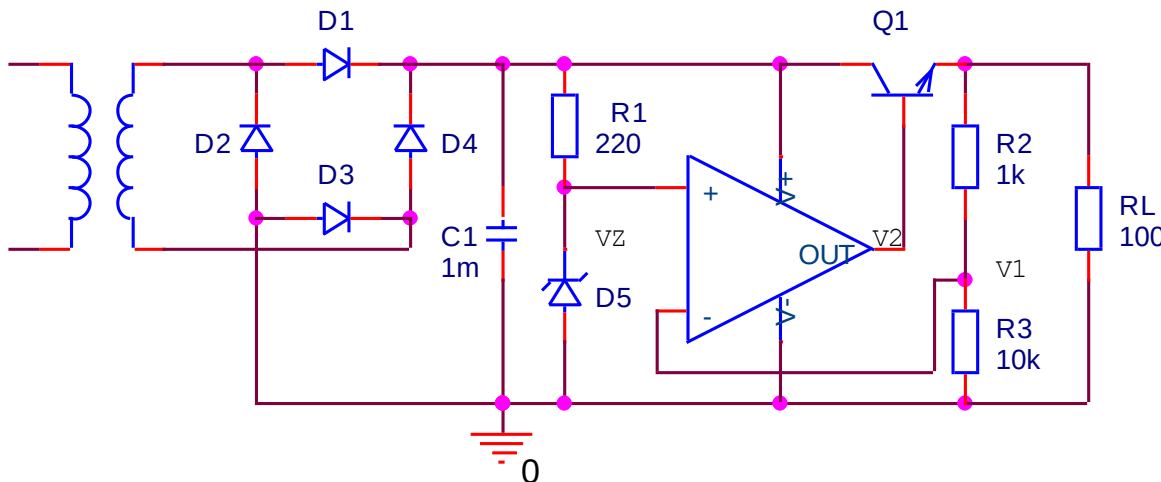


Voltage regulator with feedback and error amplifier

$$V_1 = \frac{R_3}{R_2 + R_3} V_{OUT}$$

- if V_{OUT} lower $\rightarrow V_1$ lower $\rightarrow V_2$ higher $\rightarrow V_{BE}$ higher $\rightarrow V_{CE}$ lower $\rightarrow V_{OUT}$ higher
- if V_{OUT} higher $\rightarrow V_1$ higher $\rightarrow V_2$ lower $\rightarrow V_{BE}$ lower $\rightarrow V_{CE}$ higher $\rightarrow V_{OUT}$ lower
- Because $V_1 = V_Z$

$$V_{OUT} = \left(1 + \frac{R_2}{R_3} \right) V_Z$$

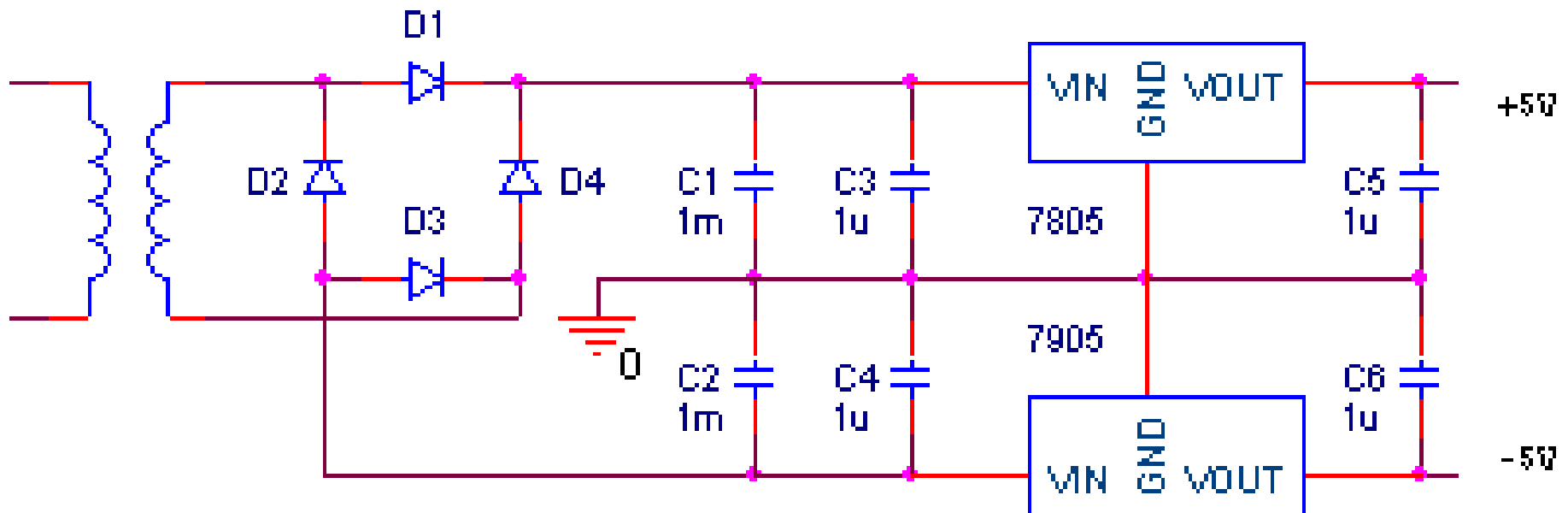


Integrated Circuits (IC Regulators)

- 7800 series provides positive output regulated voltages
- 7900 series provides negative output regulated voltages
- Last two digits from code show regulated voltage value
- Input capacitance avoids oscillations due to stray inductances of supply line
- Output capacitance acts as filter
- Usefull for ICs (ex. Arduino)

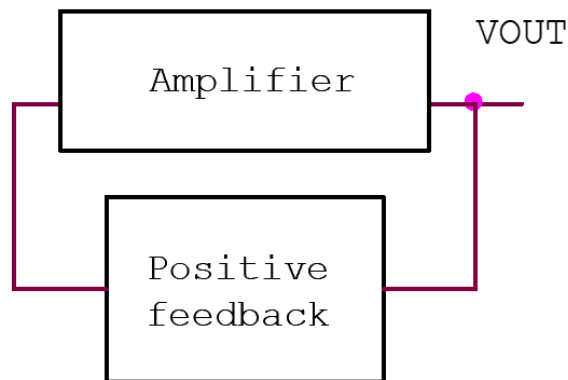
7805 = positive output regulated voltages of 5V

7905 = negative output regulated voltages of 5V



Oscillators

- Generates a repetitive waveform, without any input stimulus
- Amplifier and positive feedback
- Positive feedback: an amount of output voltage is fed back to input with zero phase shift, providing a gain of the output voltage
- Conditions for having oscillations:
 - Total phase shift must be zero
 - Voltage gain must be greater than 1 for obtaining desired oscillations amplitude. After gaining desired amplitude for oscillations, voltage amplification must stay at 1



- Used for: clock generation, carrier frequency for mobile operators, in signal generators in the lab, etc.

Works only at lower frequencies

Gain of 3

D1

D2

R2

20K

R3

20K

-

VOUT

OUT

Total Gain of 1, when the necessary amplitude is reached

R1
10K

0

+

C2

R5

2K

C1

1n

R4

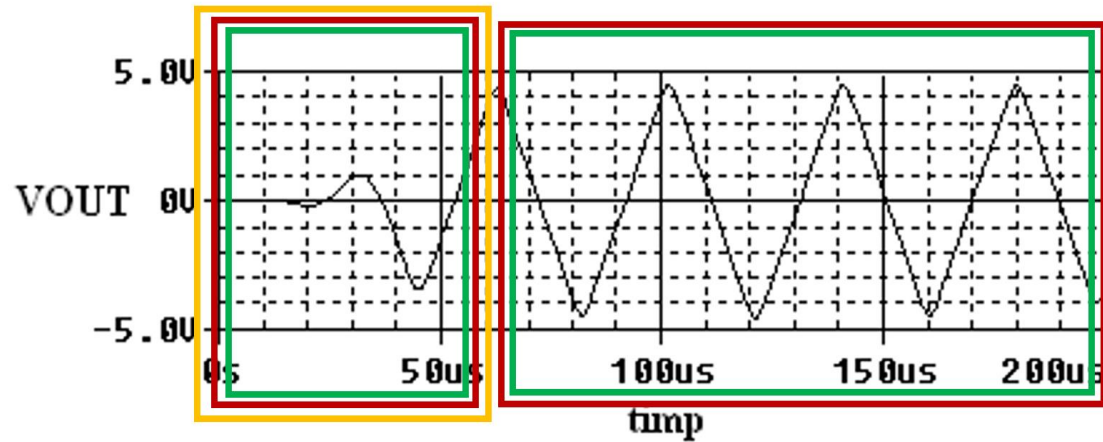
2K

0

0

Gain of 1/3

Wien bridge oscillator



Wien bridge Oscillator

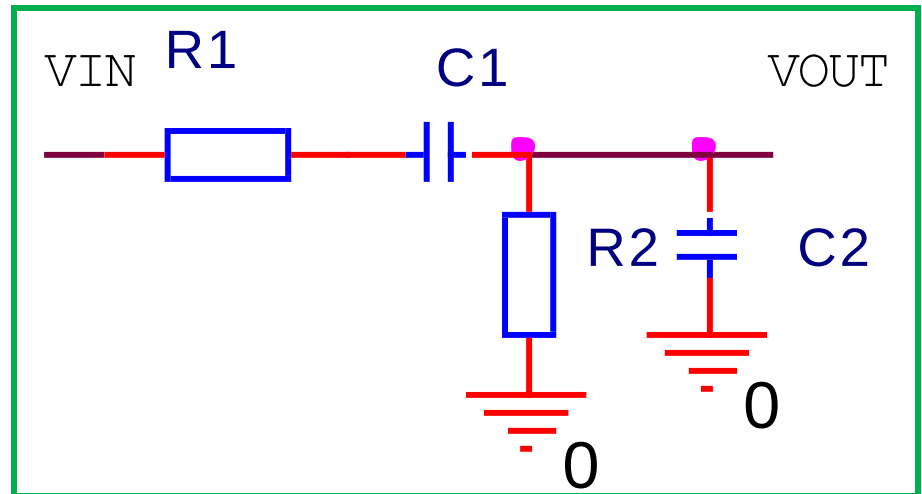
- Positive feedback circuit, made up by an op-amp and RC bridge circuit - *generate the oscillation frequency*

Wien bridge

- RC high-pass and RC low-pass filters
- Each filter operates over 'edge' frequencies (frequency adjustment)
- $R_1=R_2=R$, $C_1=C_2=C$
- attenuation is minimum for **oscillation frequency** (*Resonant Frequency f_r*)

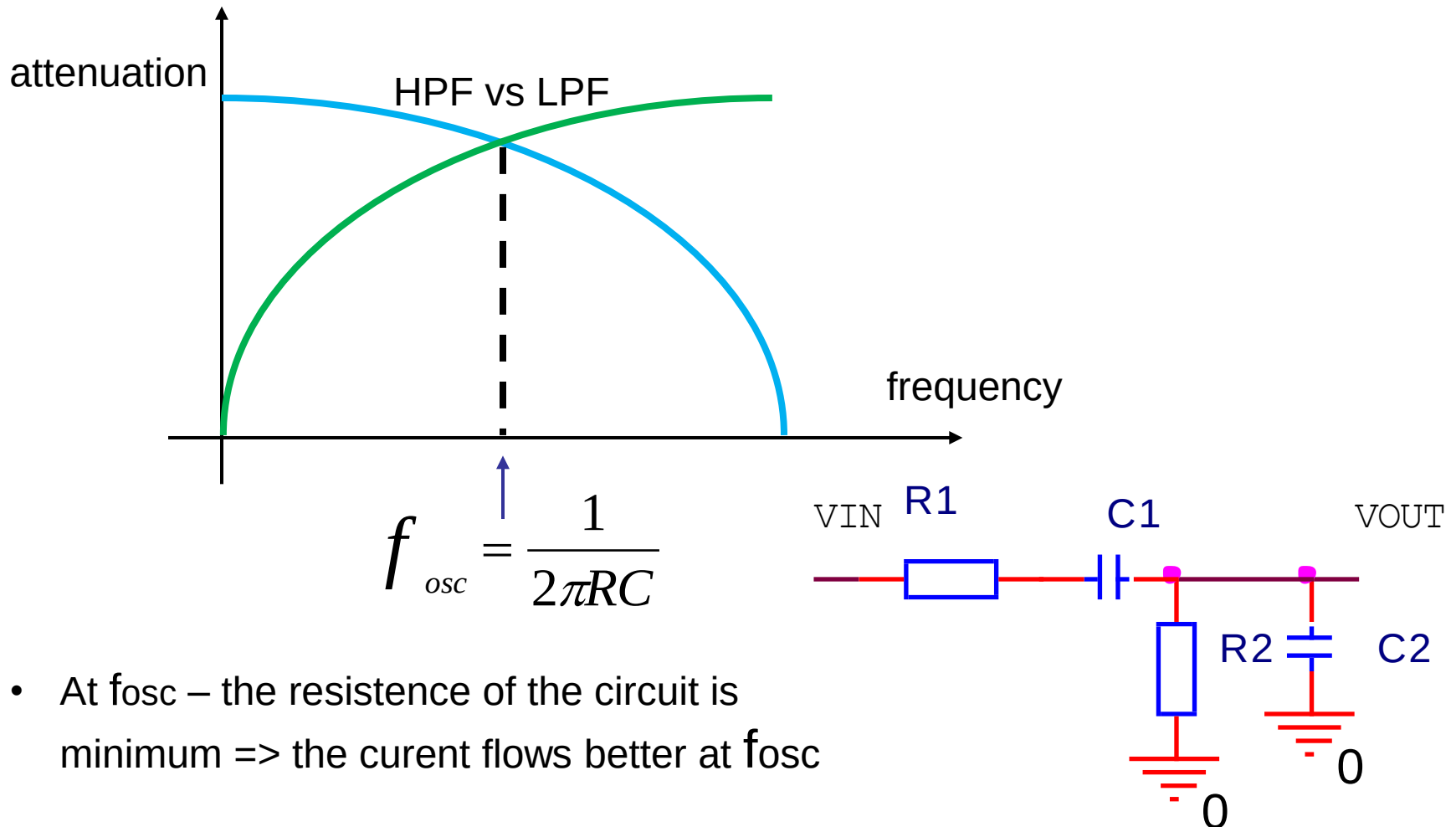
$$f_{osc} = \frac{1}{2\pi RC}$$

$$\frac{V_{IN}}{V_{OUT}} = 3 \quad \Rightarrow \text{Gain of } 1/3$$



Wien bridge Oscillator

- RC high-pass and RC low-pass filters
- attenuation is minimum for **oscillation frequency** (*Resonant Frequency f_r*)



- At f_{osc} – the resistance of the circuit is minimum => the current flows better at f_{osc}

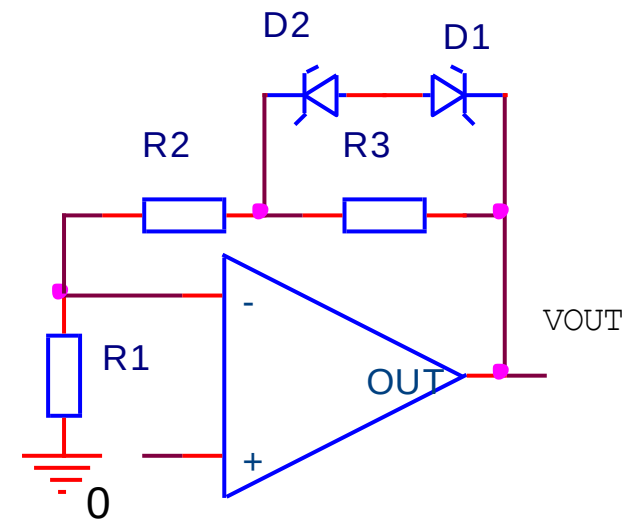
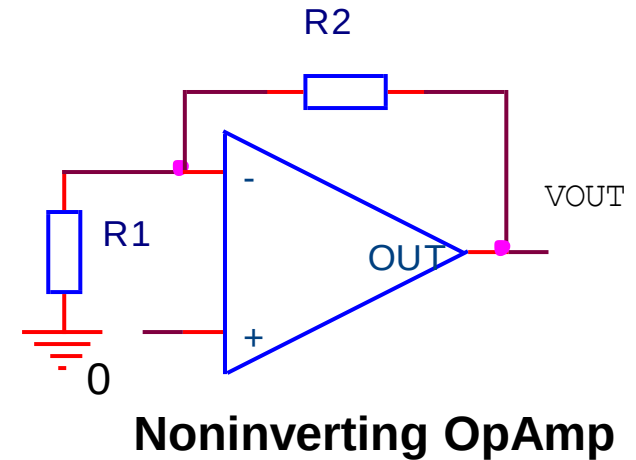
- Base circuit
 - Op-amp with negative feedback
 - For having a unit voltage gain for oscillator, amplification for base circuit must be 3

$$\frac{V_{OUT}}{V_{IN}} = 1 + \frac{R_2}{R_1} \quad R_2 = 2 R_1$$

- For lower oscillation amplitudes the two Zener diodes are off (no direct diode current)

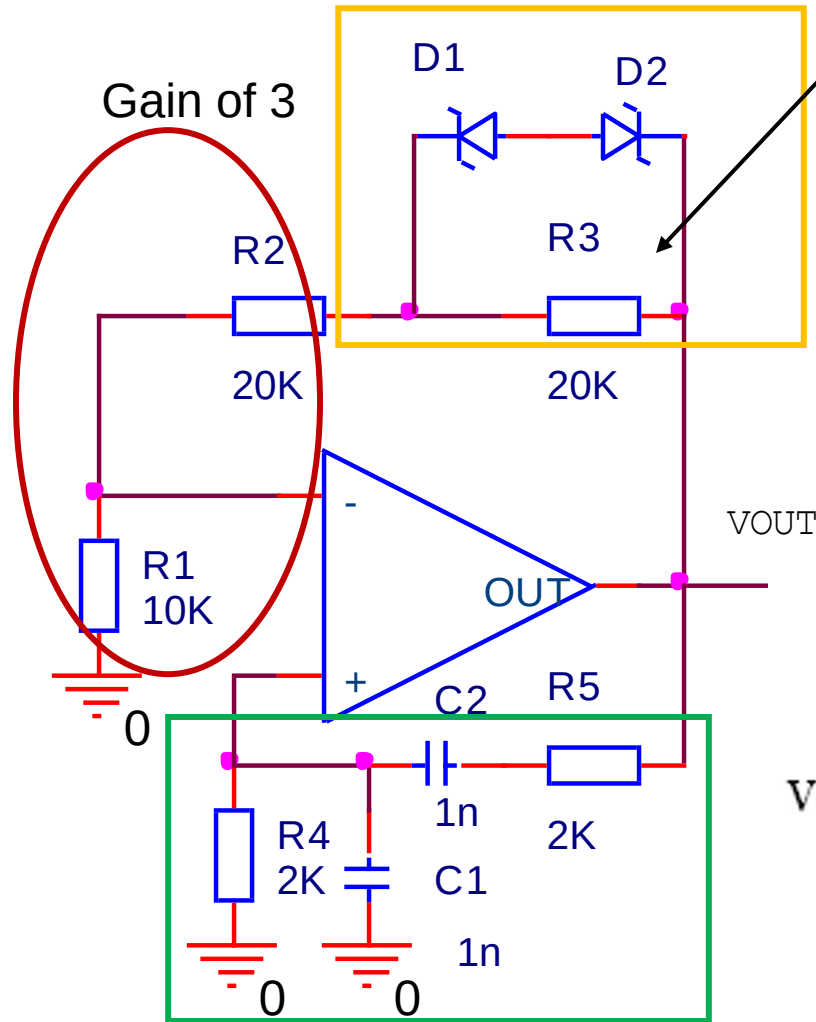
$$\frac{V_{OUT}}{V_{IN}} = 1 + \frac{R_2 + R_3}{R_1} = 3 + \frac{R_3}{R_1}$$

- When having desired amplitude, the diodes go conducting, shorting R_3 and giving a voltage gain for the base circuit of 3, and a voltage gain for oscillator of 1



Works only at lower frequencies

Total Gain of 1, when the necessary amplitude is reached



Gain of 3

D1

D2

R2

20K

R3

20K

R1
10K

VOUT

OUT

C2

R5

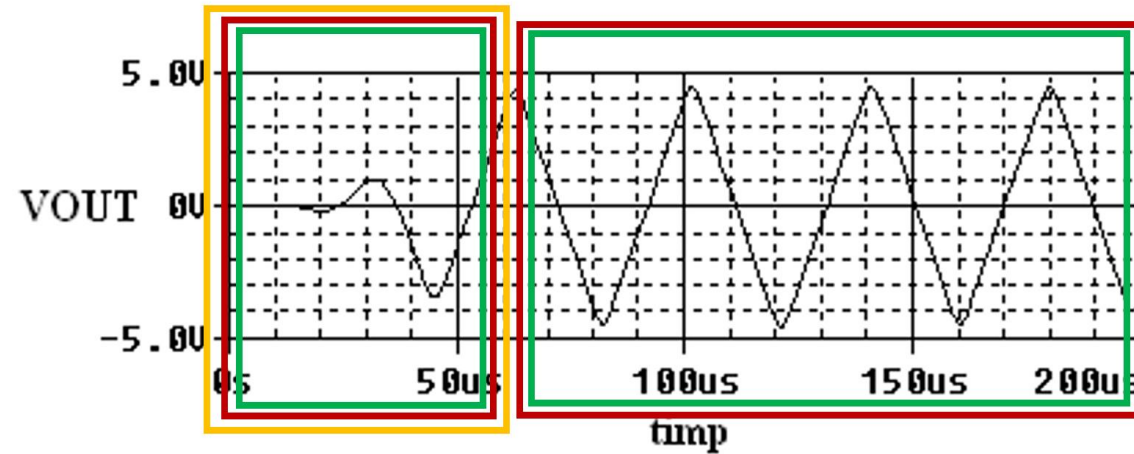
R4
2K

C1
1n

1n

2K

Gain of 1/3



Wien bridge oscillator

Proposed problems

- Design an amplifier with an amplification factor equal to -10.
- Design an amplifier with an amplification factor equal to 5.